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Information processing apparatus, control method, and program

Abstract

An information processing apparatus (2000) includes a product recognition unit (2020), a shelf label recognition unit (2040), and a determination unit (2060). The product recognition unit (2020) recognizes the number of types of products or positions of each type of products from a target image. The shelf label recognition unit (2040) recognizes the number or positions of shelf labels indicating information of the products from the target image. The determination unit (2060) determines whether or not there is an error in a relationship between the recognized number of types of products and the recognized number of shelf labels or in a relationship between the recognized positions of each type of products and the recognized positions of the shelf labels.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of U.S. patent application Ser. No. 17/675,113 filed on Feb. 18, 2022, which is a continuation application of U.S. patent application Ser. No. 16/744,947 filed on Jan. 16, 2020, which issued as U.S. Pat. No. 11,288,627, which is a continuation application of U.S. patent application Ser. No. 15/506,299 filed on Feb. 24, 2017, which issued as U.S. Pat. No. 10,579,962, which is a National Stage Entry of international application PCT/JP2015/077268, filed on Sep. 28, 2015, which claims the benefit of priority from Japanese Patent Application 2014-201363 filed on Sep. 30, 2014, the disclosures of all of which are incorporated in their entirety by reference herein.

TECHNICAL FIELD

(1) The present invention relates to an information processing apparatus, a control method, and a program.

BACKGROUND ART

(2) On a product shelving in a store, there is a shelf label near a product that describes information such as a name or price of the product. In recent years, a system for supporting management of such the shelf label is developed.

(3) Patent Document 1 discloses a system in which store shelves have store labels that display product information and shelf label identification information, and the system performs: imaging shelf label identification information of each shelf label with a camera, recognizing the imaged shelf label, and managing the shelf label. Specifically, this system generates a map representing a position of each shelf label, or detects the shelf label the expiration date for use of which has been expired and reports the detected shelf label.

RELATED DOCUMENT

Patent Document

(4) [Patent Document 1] Japanese Patent Application Publication No. 2014-48752

SUMMARY OF THE INVENTION

Technical Problem

(5) It is necessary to re-arrange, add or reduce the shelf labels according to a change of a line-up of products in a store. In addition, the position of the shelf label is sometimes shifted erroneously by a customer or a salesperson, and the position of the shelf label has to be corrected in such a case. If the customer or salesperson forgets to change the arrangement of shelf labels or makes a mistake in changing of the arrangement of shelf labels, the number or positional relationship between the shelf label and the product becomes inconsistent, and it causes a problem that the customer is not able to appropriately recognize the information of the product. The system of Patent Document 1 does not disclose a technology for managing the consistency on the number or positional relationship between the shelf label and the product.

(6) The invention has been made in view of the aforementioned problems. The present invention

provides a technology for supporting the checking of arrangement of products or shelf labels.

Solution to Problem

(7) An information processing apparatus provided by the present invention includes: a product recognition unit recognizing a number of types of products or positions of each type of products from an image on which the products and shelf labels are imaged; a shelf label recognition unit recognizing a number or positions of shelf labels indicating information of each product from the image; and a determination unit determining whether or not there is an error in a relationship between the recognized number of types of products and the recognized number of shelf labels or in a relationship between the recognized positions of each type of products and the recognized positions of the shelf labels.

(8) A control method provided by the present invention is performed by a computer. The control method includes: a product recognition step of recognizing a number of types of products or positions of each type of products from an image on which the products and shelf labels are imaged; a shelf label recognition step of recognizing a number or positions of shelf labels indicating information of the products from the image; and a determination step of determining whether or not there is an error in a relationship between the recognized number of types of products and the recognized number of shelf labels or in a relationship between the recognized positions of each type of products and the recognized positions of the shelf labels.

(9) A program provided by the present invention causes a computer to have a function of operating as the information processing apparatus provided by the present invention.

Advantageous Effects of Invention

(10) According to the present invention, a technology for supporting the checking of arrangement of products or shelf labels is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above objects, other objects, advantages and features will be more apparent from the following description of certain preferred embodiments taken in conjunction with the accompanying drawings.

(2) FIG. 1 is a block diagram illustrating an information processing apparatus according to Exemplary Embodiment 1.

(3) FIG. 2 is a flowchart illustrating a flow of processes performed by the information processing apparatus of Exemplary Embodiment 1.

(4) FIG. 3 is a block diagram illustrating a hardware configuration of a computer that realizes the information processing apparatus.

(5) FIG. 4 is a diagram illustrating a target image.

(6) FIG. 5 is a first diagram illustrating a target image representing a case where the number relationship is incorrect.

(7) FIG. 6 is a second diagram illustrating the target image representing a case where the number relationship is incorrect.

(8) FIG. 7 is a diagram illustrating a target image representing a case where the positional relationship is incorrect.

(9) FIG. 8 is a diagram illustrating a target image representing a case where a shelf label is positioned in a boundary between two product groups.

(10) FIG. 9 is a diagram illustrating a zoomed area near the boundary between the product groups of FIG. 8.

(11) FIG. 10 is a flowchart illustrating a flow of processes performed by the information processing apparatus of Exemplary Embodiment 1 in which both the number relationship and the positional

relationship are used.

(12) FIG. **11** is a block diagram illustrating an information processing apparatus according to Exemplary Embodiment 2.

(13) FIG. **12** is a flowchart illustrating a flow of processes performed by the information processing apparatus of Exemplary Embodiment 2.

DESCRIPTION OF EMBODIMENTS

(14) Hereinafter, exemplary embodiments of the present invention will be described with reference to the drawings. In all the drawings, the same components will be assigned the same reference numerals, and the description thereof will not be appropriately repeated.

Exemplary Embodiment 1

(15) FIG. **1** is a block diagram illustrating an information processing apparatus **2000** according to Exemplary Embodiment 1. In FIG. **1**, arrows represent information flows. In FIG. **1**, each block depicts not a configuration of a hardware unit but a configuration of a functional unit.

(16) The information processing apparatus **2000** includes a product recognition unit **2020**, a shelf label recognition unit **2040**, and a determination unit **2060**. The product recognition unit **2020** recognizes the number of types of products or positions of each type of products from an image (hereinafter, referred to as a target image) on which products and shelf labels are imaged.

(17) Here, “recognizing the number of types of products” means recognizing a group of products having the same type as one type when that products having the same type are placed with being adjacent to one another. For example, suppose that three columns of Products A, two columns of Products B, and two columns of Products C are sequentially placed. In this case, the number of types of products is three. Hereinafter, the group of products having the same type arranged so as to be adjacent to one another as stated above is described as a product group. Note that, the number of products included in the product group may be one.

(18) The “positions of each type of products” means positions of the product groups. More specifically, in a case where three columns of Products A, two columns of Products B, and two columns of Products C are sequentially placed, the position of the product group of Products A, the position of the product group of Products B, and the position of the product group of Product C are the “positions of each type of products”. Concrete methods of how to represent the “position of the product group” or the like will be described below.

(19) The shelf label recognition unit **2040** recognizes from the target image the number or positions of the shelf labels that indicate information of each product.

(20) The determination unit **2060** determines an error in the relationship between the number of the recognized types of products (the number of product groups) and the number of shelf labels or the relationship between the positions of each type of products (the positions of the product groups) and the shelf label.

(21) <Flow of Processes>

(22) FIG. **2** is a flowchart illustrating a flow of processes performed by the information processing apparatus **2000** of Exemplary Embodiment 1. The information processing apparatus **2000** acquires the target image (S**102**). The product recognition unit **2020** recognizes the number or positions of product groups from the target image (S**104**). The shelf label recognition unit **2040** recognizes the number or positions of shelf labels from the target image (S**106**). The determination unit **2060** determines an error in the relationship between the number of the recognized types of product groups and the number of shelf labels or the relationship between the position of the recognized product group and the position of the shelf label (S**108**).

(23) According to the information processing apparatus **2000** of the present exemplary embodiment, an error is determined in the relationship between the number of product groups (the number of the types of products) and the number of shelf labels or in the relationship between the positions of the product groups (the positions of each type of products) and the positions of the shelf labels, by using the target image on which the products and the shelf labels are imaged. Thus,

in a case where there is an error in the relationship between the number of product groups and the number of shelf labels or in the relationship between the position of the product group and the position of the shelf label, the salesperson can appropriately set the shelf label and the product group by correcting the number or position of the shelf labels or the product groups. Note that, concrete situations in which there is an error in the relationship between the number of product groups and the number of shelf labels or concrete situations in which there is an error in the relationship between the position of the product group and the position of the shelf label will be described below in conjunction with concrete examples.

(24) Hereinafter, the information processing apparatus **2000** will be described in more detail.

(25) <Hardware Configuration Example>

(26) The respective functional components of the information processing apparatus **2000** may be realized by hardware that realizes each functional component (for example, a hard-wired electronic circuit), or may be realized by a combination of hardware and software (for example, a combination of an electronic circuit and a program that controls the electronic circuit). Hereinafter, in a case where each functional component is realized by the combination of hardware and software, the configuration thereof will be illustrated in detail.

(27) The information processing apparatus **2000** may be realized by various types of computers, such as a portable terminal, a personal computer (PC), or a server. Here, the information processing apparatus **2000** may be realized by a dedicated computer for implementing the information processing apparatus **2000**, or may be realized by a general-purpose computer that operates other applications.

(28) FIG. **3** is a block diagram illustrating a hardware configuration of a computer **5000** that realizes the information processing apparatus **2000**. The computer **5000** includes a bus **5020**, a processor **5040**, a memory **5060**, a storage **5080**, and an input and output interface **5100**. The bus **5020** is a data transmission path through which the processor **5040**, the memory **5060**, the storage **5080**, and the input and output interface **5100** transmit and receive data to and from one another. Here, the method of connecting the processor **5040** and the like is not limited to bus connection. The processor **5040** is, for example, a processing device such as a central processing unit (CPU) or a graphics processing unit (GPU). The memory **5060** is, for example, a memory such as a random access memory (RAM) or a read only memory (ROM). The storage **5080** is, for example, a storage apparatus such as a hard disk, a solid state drive (SSD), or a memory card. The storage **5080** may be a memory such as RAM or ROM.

(29) The input and output interface **5100** is an input and output interface through which the computer **5000** transmits and receives data to and from an external apparatus. For example, in a case where the information processing apparatus **2000** acquires the target image from the external apparatus, the computer **5000** that realizes the information processing apparatus **2000** is connected to the external apparatus through the input and output interface **5100**. Note that, there are various methods of connecting the computer **5000** to the external apparatus through the input and output interface. For example, such connection is bus connection using a bus line (for example, universal serial bus (USB)) or network connection using a network line. Note that the network line may be a wireless line, or may be a wired line.

(30) The storage **5080** stores a program for realizing the function of the information processing apparatus **2000**. Specifically, program modules for respectively realizing the functions of the product recognition unit **2020**, the shelf label recognition unit **2040**, and the determination unit **2060** are stored. The processor **5040** realizes the functions of the product recognition unit **2020**, the shelf label recognition unit **2040**, and the determination unit **2060** by executing the respective program modules. Here, when the respective modules are executed, the processor **5040** may read these modules into the memory **5060** and may execute the read module, or may execute the module without reading the module into the memory **5060**.

(31) Note that, the hardware configuration of the computer **5000** used for realizing the information

processing apparatus **2000** is not limited to the configuration illustrated in FIG. 3. For example, the respective program module may be stored in the memory **5060**. In this case, the computer **5000** may not include the storage **5080**.

(32) <Acquisition Method of Target Image>

(33) The target image is imaged by a camera provided in a store. Only one type of product or one shelf label may be pictured or a plurality of different products or a plurality of shelf labels may be pictured on one target image.

(34) For example, the information processing apparatus **2000** acquires the target image stored within this camera. In a case where an external storage device is stored in the target image imaged by the camera, the information processing apparatus **2000** may acquire the target image from this storage device. The camera may be provided such that the imaged target image is stored in a storage device provided within the information processing apparatus **2000**. In this case, the information processing apparatus **2000** acquires the target image from a storage unit provided within the information processing apparatus **2000**.

(35) The camera may be provided integrally with the computer that realizes the information processing apparatus **2000**, or may be provided outside the computer. As an example in which the camera is provided integrally with the computer that realizes the information processing apparatus **2000**, the information processing apparatus **2000** may be realized by a portable terminal equipped with the camera. In this case, the information processing apparatus **2000** performs a process on the target image imaged by the camera of the portable terminal.

(36) <Details of Product Recognition Unit **2020**>

(37) The product recognition unit **2020** analyzes the target image, and recognizes a product. Here, a technique for recognizing an object such as a product pictured on an image is a well-known technique, and thus various well-known techniques may be used for recognizing the object. Hereinafter, an example of a process performed by the product recognition unit **2020** will be described.

(38) FIG. 4 is a diagram illustrating the target image **10**. A product shelving on which the products are exhibited is pictured on the target image **10**. Note that, the way of exhibiting the products is not limited to exhibit the products on the product shelving, as long as the products and the shelf labels may be pictured on the target image.

(39) An outer frame **20** is an outer frame of the product shelving. A shelf plate **30** is a plate on which the products are placed. A product group **40** represents a product group. Three product groups are pictured on the target image **10** of FIG. 4. A shelf label **50** is a label displaying information of the product. In FIG. 4, three shelf labels are pictured on the target image **10**. Here, the shelf label **50** may be a shelf label created as an analog medium such as paper, or may be a shelf label created as an electronic medium.

(40) Initially, the product recognition unit **2020** recognizes the product from the target image **10**. The product recognition unit **2020** recognizes a group of products among the recognized products that are adjacent to each other and have similar feature values, as one product group **40**. For example, the product group **40-1**, the product group **40-2**, and the product group **40-3** are recognized from the target image **10** in FIG. 4.

(41) <<Method 1 of Recognizing Product Group **40**>>

(42) For example, the product recognition unit **2020** classifies the respective products into the product groups by sequentially comparing the feature values of the adjacent products from any of the left and the right. Specifically, the product recognition unit **2020** classifies these products into the same product group **40** if the feature values of the adjacent products are the same as or similar to each other. On the other hand, the product recognition unit **2020** classifies these products into different product groups **40** if the feature values of the adjacent products are not similar to each other.

(43) For example, in the example of FIG. 4, the product recognition unit **2020** sequentially

compares the feature values of the respective products placed on a shelf plate from the left. Initially, the product recognition unit **2020** determines that the feature values of the first product and the second product are similar to each other, and classifies the first product and the second product into the same product group **40-1**. Similarly, the product recognition unit **2020** determines that the feature values of the second product and the third product are similar to each other, and also classifies the third product into the same product group **40-1**. On the other hand, as a result acquired by comparing the feature values of the third product and the fourth product, the product recognition unit **2020** determines that these feature values are not similar. As a result, the fourth product is classified into the product group **40-2**, which is a product group different from the product group **40-1**.

(44) According to this method, the product recognition unit **2020** does not need to recognize the feature values of the respective products in advance. Thus, it is possible to reduce computer resources required to realize the information processing apparatus **2000**.

(45) <<Method 2 of Recognizing Product Group **40**>>

(46) The product recognition unit **2020** may recognize the product group **40** by using information (hereinafter, referred to as product information) that is acquired by associating the product with a feature value describing a feature of the appearance of the product. The product information may be stored in the storage unit provided inside or outside the information processing apparatus **2000** in advance. The product recognition unit **2020** acquires the product information from the storage unit.

(47) As for the product information, one or a plurality of feature values may be associated with one product. Here, in a case where the plurality of features is associated with one product, these feature values are, for example, feature values indicating the features of the product viewed in different directions. In some products, the appearance of the product looks very different depending on directions in which the product is viewed. For example, a certain product would have patterns on the label attached thereto that look very different between when being viewed in a certain direction and when being viewed in another direction. Since the exhibited products are not necessarily placed to face to the same direction, when the appearance of the product looks very different depending on the directions in which the product is viewed, the feature values of the product extracted from the target image may be different even though the same product is viewed.

(48) Thus, the product recognition unit **2020** determines the product corresponding to the calculated feature values, by comparing the feature values calculated from the target image **10** with the feature value indicated by the product information. The product recognition unit **2020** classifies the products that are the same product as each other and are adjacent to each other into the same product group **40**.

(49) According to this method, the same type of products (products placed in different directions), the feature values calculated from the target image of which are not similar to each other, are classified into the same product group **40**. Thus, it is possible to increase the accuracy of the classification of the product group **40**.

(50) <<Method of Recognizing Number of Product Group **40**>>

(51) The product recognition unit **2020** recognizes the number of the product group **40** by counting the number of recognized product groups **40**.

(52) <<Method of Recognizing Position of Product Group **40**>>

(53) For example, the product recognition unit **2020** recognizes a coordinate of each product group **40** on the target image **10**, as the position of the product group **40**. Here, the coordinate of the product group **40** is a coordinate satisfying a predetermined condition within an area indicating the product group **40**. For example, this condition is a center, a left end, or a right end of the area indicating the product group **40**. Note that, the product recognition unit **2020** may represent the position of the product group **40** as a two-dimensional coordinate (a combination of a horizontal direction with a vertical direction), or may represent the position thereof as a one-dimensional

coordinate (a coordinate in the horizontal direction).

(54) <Details of Shelf Label Recognition Unit **2040**>

(55) The shelf label recognition unit **2040** analyzes a target image, and recognizes a shelf label. Here, similarly to the product recognition unit **2020**, the shelf label recognition unit **2040** can use various existing techniques for recognizing an object from the image. Hereinafter, an example of a process performed by the shelf label recognition unit **2040** will be described.

(56) <<Recognition Method of Shelf Label **50**>>

(57) Similarly to the case of the product recognition unit **2020**, the recognition method of the shelf label will be described with reference to FIG. **4**. For example, the shelf label recognition unit **2040** recognizes the shelf label **50** by acquiring information indicating a feature value of the shelf label **50** and recognizing an object matching this feature value from the target image. Here, the feature value of the shelf label **50** may be stored in the storage unit provided inside or outside the information processing apparatus **2000** in advance.

(58) For example, the shelf label recognition unit **2040** may recognize the area indicating the shelf plate **30**, and then recognize as the shelf label **50** an object having a predetermined shape (a rectangle or the like) included in this area indicating the shelf plate **30**. For example, the shelf label recognition unit **2040** recognizes the shelf plate by acquiring the feature value indicating the shelf plate and using the acquired feature value. Here, the feature value of the shelf plate **30** or the predetermined shape indicating the shelf label **50** may be stored in the storage unit provided inside or outside the information processing apparatus **2000** in advance.

(59) <<Method of Recognizing Number of Shelf Label>

(60) The shelf label recognition unit **2040** recognizes the number of shelf labels **50** by counting the number of shelf labels **50** recognized by the aforementioned method.

(61) <<Method of Recognizing Position of Shelf Label>

(62) The method of causing the shelf label recognition unit **2040** to recognize the position of the shelf label **50** is the same as the method of causing the product recognition unit **2020** to recognize the position of the product group **40**.

(63) <Details of Determination Unit **2060**>

(64) As stated above, the determination unit **2060** determines an error in the relationship between the number of recognized product groups **40** and the number of shelf labels **50**, or the relationship between the position of the product group **40** and the position of the shelf label **50**.

(65) <<Determination Method of Number Relationship>>

(66) The determination unit **2060** compares the number of product groups **40** with the number of shelf labels **50**. For example, the determination unit **2060** determines that there is an error in the relationship between the numbers if the number of product groups **40** and the number of shelf labels are different.

(67) The determination method will be described in detail with reference to FIGS. **4** to **6**. In the aforementioned example of FIG. **4**, the number of product groups **40** is 3, and the number of shelf labels **50** is 3. Thus, the determination unit **2060** determines that the relationship between the number of product groups **40** and the number of the shelf labels **50** is correct.

(68) FIG. **5** is a first diagram illustrating the target image representing a case where the number relationship is incorrect. In the aforementioned example of FIG. **5**, the number of product groups **40** is 3, whereas the number of shelf labels **50** is 2. Thus, the determination unit **2060** determines that there is an error in the relationship between the number of product groups **40** and the number of shelf labels **50**. For example, the situation illustrated by FIG. **5** may occur when the salesperson forgets an operation of adding a shelf label **50-3** even though the product in the product group **40-3** is newly exhibited or when the shelf label **50-3** falls off. The salesperson can recognize that such a situation occurs, by using the result of the determination performed by the determination unit **2060**.

(69) FIG. **6** is a second diagram illustrating the target image representing a case where the number relationship is incorrect. In the example of FIG. **6**, the number of product groups **40** is 2, whereas

the number of shelf labels **50** is 3. In this case, the determination unit **2060** determines that there is an error in the relationship between the number of product groups **40** and the number of shelf labels **50**. For example, the situation illustrated by FIG. **6** may occur when the salesperson forgets removing the shelf label **50-3** even though the exhibition of the product of the product group is terminated, when the salesperson forgets adding the product even though the shelf label **50-3** is added in order to newly exhibit the product of the product group **40-3**, or the product of the product group **40-3** is sold out. The determination result performed by the determination unit **2060** is used, and thus, the salesperson can recognize that such a status occurs.

(70) <<Determination Method of Positional Relationship>>

(71) For example, if there is not the shelf label **50** in a predetermined direction to the product group **40**, the determination unit **2060** determines that there is an error in the relationship between the position of the product group **40** and the position of the shelf label **50**. For example, the predetermined direction is up, down, left, or right.

(72) Here, as described above, suppose that information indicating a direction in which the shelf label **50** is attached for the product group **40** is previously stored in the storage unit inside or outside the information processing apparatus **2000**. For example, in a case where the shelf label **50** is attached above the product group **40**, if there is no shelf label **50** above the product group **40**, the determination unit **2060** determines that there is an error in the relationship between the position of the product group **40** and the position of the shelf label **50**.

(73) The process performed by the determination unit **2060** will be described in detail with reference to FIGS. **4** and **7**. In these diagrams, it is assumed that the shelf label **50** is attached under the product group **40**.

(74) In FIG. **4**, a shelf label **50-1** is located under the product group **40-1**, a shelf label **50-2** is located under the product group **40-2**, and a shelf label **50-3** is located under the product group **40-3**. Thus, the determination unit **2060** determines that the relationship between the position of product groups **40** and the position of the shelf labels **50** is correct.

(75) FIG. **7** is a diagram illustrating a target image representing a case where the positional relationship is incorrect. In FIG. **7**, the shelf label **50** is not located under the product group **40-1**. Thus, the determination unit **2060** determines that there is an error in the relationship between the position of product groups **40** and the position of shelf labels **50**.

(76) Here, there would be a case in which the shelf label is positioned in a boundary between two product groups, and the same shelf label is positioned under the two product groups. FIG. **8** is a diagram illustrating a target image representing a case where a shelf label is positioned in a boundary between two product groups. In FIG. **8**, the shelf label **50** is positioned in a boundary between two product groups **40-1** and **40-2**. The determination unit **2060** needs to determine under which product group the shelf label is located.

(77) In such a case, the determination result performed by the determination unit **2060** could be different based on under which product group **40** the shelf label **50** is determined to be located. If it is determined that the shelf label **50** is located under the product group **40-1**, the shelf label **50** is attached under each product group **40**. Thus, the determination unit **2060** determines that the relationship between the position of product groups **40** and the position of the shelf labels **50** is correct. On the other hand, if it is determined that the shelf label **50** is located under the product group **40-2**, there is no shelf label **50** under the product group **40-1**. Thus, the determination unit **2060** determines that there is an error in the relationship between the position of product groups **40** and the position of shelf labels **50**.

(78) For example, when there is the shelf label **50** in the boundary between two product groups **40** as stated above, the determination unit **2060** determines that the shelf label **50** is located under the closer product group **40**. The determination method will be described in detail with reference to FIG. **9**. FIG. **9** is a diagram illustrated by zooming an area near the boundary between the product group and the product group **40-2** of FIG. **8**. In order to easily view the diagram, the described

content of the shelf label **50-1** will not be repeated. In FIG. 9, a horizontal distance between the right end of the product group **40-1** and the center of the shelf label **50-1** is $d1$. In FIG. 9, a horizontal distance between the left end of the product group **40-2** and the center of the shelf label **50-1** is $d2$. Here, since $d2$ is less than $d1$, the product group **40-2** is located closer to the shelf label **50-1** than the product group **40-1** is. Thus, the determination unit **2060** determines that the shelf label is located under the product group **40-2**.

(79) As a result, in the example of FIG. 8, there is no shelf label **50** under the product group **40-1**. Thus, the determination unit **2060** determines that there is an error in the relationship between the position of product groups **40** and the position of shelf labels **50**.

(80) Note that, in a case where a plurality of rows of the product shelving is pictured on the target image, the determination unit **2060** may take into consideration which product group **40** and which shelf label **50** are included in the same row. Specifically, the determination unit **2060** limits the target of the determination of whether or not there is an error in the positional relationship or the number relationship to the products and the shelf labels included in the same row.

(81) Although it has been described in FIG. 4 that the product group **40** corresponding to the shelf label **50** is placed above the shelf label **50**, an exhibition method of placing the product group **40** corresponding to the shelf label under this shelf label **50** is also used. That is, a certain row of the product shelving becomes 1) an area including the shelf plate **30** and the product group **40** placed on the shelf plate **30** or 2) an area including the shelf plate **30** and the product group **40** placed under the shelf plate **30**. Whether the determination unit **2060** recognizes 1) or 2) as the row is set in advance. For example, information for determining whether the determination unit recognizes 1) or 2) as the row (information indicating whether the shelf label **50** is attached above or under the product group **40**) is stored in advance in the storage unit provided inside or outside the information processing apparatus **2000**. The product recognition unit **2020** determines an area recognized as one row by referring to this information. The product recognition unit **2020** determines the relationship between the product group **40** and the shelf label **50** included in the same row.

(82) The determination unit **2060** may determine that there is an error in the relationship between the position of the product group **40** and the position of the shelf label **50** in a case where there is no shelf label **50** within a predetermined distance range from a certain product group **40**. In this case, it is assumed that the predetermined distance is stored in the storage unit outside or inside the information processing apparatus **2000**. The determination unit **2060** may take both the direction and the distance into consideration. Specifically, the determination unit **2060** determines that there is an error in the positional relationship between the product group **40** and the shelf label **50** in a case where there is no shelf label **50** that satisfies the condition in which “a certain product group **40** is located in a predetermined direction with respect to the product group **40** and a distance from the product group **40** is equal to or less than a predetermined distance”.

(83) <Determination Method of Focusing on Both Number Relationship and Positional Relationship>

(84) The determination unit **2060** may determine an error in the relationship between the product group **40** and the shelf label **50** by focusing on both the number relationship and the positional relationship. The determination method will be described in detail with reference to FIG. 5. In the situation of FIG. 5, if the determination unit **2060** determines an error in the relationship between the position of the product group **40** and the position of the shelf label **50**, it is determined that there is an error in the positional relationship. The shelf label **50** is not located under the product group **40-3**. However, in some points of view, the situation of FIG. 5 may mean that “there is no error in the positional relationship and there is an error in the number relationship”.

(85) Thus, the determination unit **2060** may determine whether or not there is an error in the relationship between the position of the product group **40** and the position of the shelf label **50**, only when the relationship between the number of product groups **40** and the number of shelf labels

50 is correct. A flowchart for illustrating a flow of processes is illustrated in FIG. **10**.

(86) The information processing apparatus **2000** acquires the target image (S202). The product recognition unit **2020** recognizes the number or positions of product groups **40** from the target image (S204). The shelf label recognition unit **2040** recognizes the number or positions of shelf labels **50** from the target image (S206). The determination unit **2060** compares the number of product groups **40** with the number of shelf labels **50** (S208).

(87) In a case where the number of product groups **40** and the number of shelf labels **50** are not equal to each other (S210: NO), the process of FIG. **10** proceeds to step S212. In step S212, the determination unit **2060** determines that there is an error in the number relationship between the product groups **40** and the shelf labels **50**.

(88) On the other hand, in a case where the number of product groups **40** and the number of shelf labels **50** are equal to each other (S210: YES), the process of FIG. **10** proceeds to step S214. In step S214, the determination unit **2060** compares the position of the product groups **40** with the position of the shelf labels **50**. In a case where there is a product group **40** that has no corresponding shelf label **50** (S216: YES), the process of FIG. **10** proceeds to step S218. In step S218, the determination unit **2060** determines that there is an error in the positional relationship between the product groups **40** and the shelf labels **50**. On the other hand, in a case where there is no product group **40** that has no corresponding shelf label **50** (S216: NO), the process of FIG. **10** is ended.

(89) The determination is performed using both the number relationship and the positional relationship as stated above, and the determination result performed by the determination unit **2060** becomes more accurate.

(90) <Timing when Information Processing Apparatus **2000** Performs Processes>

(91) There are various timings when the information processing apparatus **2000** performs the processes. For example, the information processing apparatus **2000** periodically performs the processes. In this case, a date and time or a cycle in which the process is performed is set in advance for the information processing apparatus **2000**. For example, the information processing apparatus **2000** may perform the process when the operation by the salesperson is received. The information processing apparatus **2000** may perform the process when the target image is generated or when the target image is acquired.

Exemplary Embodiment 2

(92) FIG. **11** is a block diagram illustrating an information processing apparatus **2000** according to Exemplary Embodiment 2. In FIG. **11**, arrows represent information flows. In FIG. **11**, each block does not represent a configuration of a hardware unit but a configuration of a functional unit.

(93) The information processing apparatus **2000** of Exemplary Embodiment 2 includes a notification unit **2080**. The notification unit **2080** performs the notification or the displaying in a case where it is determined that there is an error by the determination unit **2060**. By performing the notification or the displaying as stated above, the salesperson can easily recognize the determination result performed by the determination unit **2060**.

(94) <Flow of Processes>

(95) FIG. **12** is a flowchart illustrating a flow of processes performed by the information processing apparatus **2000** of Exemplary Embodiment 2. Here, steps S102 to S108 are the same as those of the flowchart of FIG. **2**, and thus, the description thereof will not be repeated. The process of FIG. **12** is branched at step S302. Specifically, in a case where it is determined that there is no error in the number or positional relationship between the product groups **40** and the shelf labels **50** in step S108 (step S302: NO), the process of FIG. **12** is ended. On the other hand, in a case where it is determined that there is the error in the number or positional relationship between the product groups **40** and the shelf labels **50** in step S108 (step S302: YES), the process of FIG. **12** proceeds to step S304. In step S304, the notification unit **2080** performs the notification or displaying indicating that there is the error in the number or positional relationship between the product groups **40** and the shelf labels **50**.

(96) <About Target Terminal of Notification or Display>

(97) The salesperson present in a store into which the information processing apparatus **2000** is introduced may directly use the information processing apparatus **2000** or may use another terminal (hereinafter, referred to as a user terminal) connected to the information processing apparatus **2000** in a communicable fashion. For example, the user terminal is a portable terminal carried by each salesperson.

(98) For example, in a case where the salesperson directly uses the information processing apparatus **2000**, when the determination unit **2060** determines that there is the error in the number or positional relationship between the product groups **40** and the shelf labels **50**, the notification unit **2080** displays the notification indicating that there is the error, on the display screen connected to the information processing apparatus **2000**. Meanwhile, in a case where the salesperson uses the user terminal, when it is determined that there is the error in the number or positional relationship between the product groups **40** and the shelf labels **50**, the information processing apparatus **2000** notifies the user terminal of the determination result performed by the determination unit **2060**. The user terminal that receives this notification displays the determination result on the display screen included in the user terminal. Here, the notification unit **2080** may perform both the displaying on the information processing apparatus **2000** and the notification to the user terminal.

(99) Here, the process performed by the information processing apparatus **2000** or the user terminal in a case where it is determined that there is the error in the number or positional relationship between the product groups **40** and the shelf labels **50** is not limited to the output to the display screen. For example, the information processing apparatus **2000** or the user terminal may output voice for reporting that it is determined that there is the error, or may perform a process of turning on the light.

(100) <Notification Method of Error Content>

(101) The notification unit **2080** may notify of or display the “notification indicating that there is the error” or may notify of or display the “notification indicating the kind of the error”. For example, the notification unit **2080** notifies of information for determining a location (product shelving or row) in which it is determined that there is the error. Accordingly, the salesperson can easily recognize a location in which it is necessary to add or remove the products or the shelf labels. Note that, the notification unit **2080** performs the displaying on the display screen by the same method as various notification methods to be described below.

(102) <<Notification Method of Target Location>>

(103) For example, the notification unit **2080** performs notification including the target image on which a target location of notification (location in which the error is found in the relationship between the product groups **40** and the shelf labels **50**) is pictured. Here, in a case where a plurality of locations (product shelving) in which the products are exhibited is pictured on the target image, the notification unit **2080** may attach the image on which the target location is highlighted or may attach the image on which only the target location is cropped.

(104) For example, the notification unit **2080** may determine the location as the notification target, and may notify information related to this location. For example, this information is an ID of the product shelving or a position within the store.

(105) In a case where the ID of the product shelving is displayed on the product shelving, the notification unit **2080** recognizes the ID of the product shelving as the notification target that is pictured on the target image. The notification unit **2080** notifies of the recognized ID of the product shelving. The notification unit **2080** may determine the position of the product shelving within the store by using the ID of the product shelving, and may notify of this position. In this case, the storage unit inside or outside the information processing apparatus **2000** stores information that associates the ID of the product shelving with the position of the product shelving within the store that has this ID (hereinafter, product shelving information).

(106) The notification unit **2080** may determine the position of the product shelving within the

store as the notification target and may notify of this position by using information related to the arrangement of the product shelving (store map) and information related to the imaging range of the camera with which the target image is imaged. Here, the information related to the imaging range of the camera that images the target image may be acquired from the camera, or may be acquired from the storage unit that stores the information related to the camera. Here, the notification unit **2080** may determine the ID of the product shelving from the position of the product shelving within the store as the notification target within the store and may notify of this ID by further using the aforementioned product shelving information.

(107) The notification unit **2080** may notify of the information related to the row (row in which the error is found in the relationship between the product groups **40** and the shelf labels **50**) as the notification target by the same method as the method of notifying of the information related to the aforementioned product shelving.

(108) <Notification Method of Product Group **40** being Notification Target>

(109) For example, in a case where there is no shelf label above or under the product group **40**, the notification unit **2080** preferably notifies that there is no shelf label **50** corresponding to the product group **40**. Here, there are various methods of notifying of the product group **40** as this notification target. For example, the notification unit **2080** may perform the notification including the image on which the product group **40** of the notification target is highlighted or the image on which only the product group **40** as the notification target is cropped from the target image.

(110) For example, the notification unit **2080** determines the product group **40** as the notification target by using the aforementioned product information (information in which the product is associated with the feature value indicating the appearance feature of the product). Here, the feature value of the product of each product group **40** is already calculated by the product recognition unit **2020**. Thus, the notification unit **2080** determines the product corresponding to the feature value of the product of the product group **40** as the notification target by using the product information. For example, the notification unit **2080** notifies of the ID of the determined product group **40**.

(111) An ID for identifying the product may be assigned to the product. This ID may be represented by the character string such as alphabetical or alphanumeric characters, or may be represented by an image such as barcode or two-dimensional code. This ID is identified by the product recognition unit **2020**. In this case, the notification unit **2080** notifies of the ID of the product of the notification target.

(112) <<Determination Method of Shelf Label **50** of Notification Target>>

(113) For example, in a case where there is no product group above or under the shelf label **50**, the notification unit **2080** preferably notifies that there is no product corresponding to the shelf label **50**. For example, the notification unit **2080** may perform the notification including the image on which the shelf label **50** of the notification target is highlighted or the image on which only the shelf label **50** of the notification target is cropped from the target image.

(114) The ID may be assigned to the shelf label by the same method as the aforementioned method for assigning the ID to the product. This ID is identified by the shelf label recognition unit **2040**. In this case, the notification unit **2080** notifies of the ID of the shelf label of the notification target.

(115) <Notification Indicating that there is No Error>

(116) In a case where it is determined that there is no error in the relationship between the product groups **40** and the shelf labels **50**, the notification unit **2080** may notify or display that there is no error. In a case where it is determined that there is no error, the notification unit **2080** may notify of or display the information related to the location, the product group or the shelf label in which it is determined that there is no error.

(117) Although the exemplary embodiments of the present invention have been described with reference to the drawings, these exemplary embodiments are examples of the present invention, and the combinations of the exemplary embodiments and various configurations other than the exemplary embodiments may be adopted.

(118) Hereinafter, reference examples are appended. 1. An information processing apparatus including: a product recognition unit recognizing a number of types of products or positions of each type of products from an image on which the products and shelf labels are imaged; a shelf label recognition unit recognizing a number or positions of shelf labels indicating information of the products from the image; and a determination unit determining whether or not there is an error in a relationship between the recognized number of types of products and the recognized number of shelf labels or in a relationship between the recognized positions of each type of products and the recognized positions of the shelf labels. 2. The information processing apparatus according to 1, wherein, in a case where the recognized number of types of products and the recognized number of shelf labels are not equal to each other, the determination unit determines that there is an error in the relationship between the recognized number of types of products and the recognized number of shelf labels. 3. The information processing apparatus according to 1 or 2, wherein, in a case where the shelf label is not attached for a predetermined type of products in a predetermined direction, the determination unit determines that there is an error in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels. 4. The information processing apparatus according to any one of 1 to 3, wherein, in a case where the shelf label is not attached within a predetermined distance range from a predetermined type of products, the determination unit determines that there is an error in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels. 5. The information processing apparatus according to any one of 1 to 4, wherein, in a case where it is determined that there is an error in the relationship between the recognized number of types of products and the recognized number of shelf labels, the determination unit determines whether or not there is an error in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels. 6. The information processing apparatus according to any one of 1 to 5, further including: a notification unit performing notification or displaying in a case where the determination unit determines that there is an error in the relationship between the recognized number of types of products and the number of shelf labels or in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels. 7. A control method performed by a computer, the method including: a product recognition step of recognizing a number of types of products or positions of each type of products from an image on which the products and shelf labels are imaged; a shelf label recognition step of recognizing a number or positions of shelf labels indicating information of each product from the image; and a determination step of determining whether or not there is an error in a relationship between the recognized number of types of products and the recognized number of shelf labels or in a relationship between the recognized positions of each type of products and the recognized positions of the shelf labels. 8. The control method according to 7, wherein, in a case where the recognized number of types of products and the recognized number of shelf labels are not equal to each other, it is determined that there is an error in the relationship between the recognized number of types of products and the recognized number of shelf labels in the determination step. 9. The control method according to 7 or 8, wherein, in a case where the shelf label is not attached for a predetermined type of products in a predetermined direction, it is determined that there is an error in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels in the determination step. 10. The control method according to any one of 7 to 9, wherein, in a case where the shelf label is not attached within a predetermined distance range from a predetermined type of products, it is determined that there is an error in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels in the determination step. 11. The control method according to any one of 7 to 10, wherein, in a case where it is determined that there is an error in the relationship between the recognized number of types of products and the recognized number of shelf labels, it is determined whether or not there is an error in the relationship between the recognized positions of each type of

products and the recognized positions of the shelf labels in the determination step. 12. The control method according to any one of 7 to 11, further including: a notification step of performing notification or displaying in a case where it is determined that there is an error in the relationship between the recognized number of types of products and the number of shelf labels or in the relationship between the recognized positions of each type of products and the recognized positions of the shelf labels in the determination step. 13. A program causing a computer to operate as the information processing apparatus according to any one of claims 1 to 6.

(119) The present application claims priority based on Japanese Patent Application No. 2014-201363 filed on Sep. 30, 2014, and the entire content thereof is incorporated by reference in the present application.

Claims

1. An information processing apparatus comprising: at least one memory storing instructions; and at least one processor configured to execute the instructions to: recognize a first position of a first product and a second position of a second product from an image on which products and a shelf label are imaged by obtaining a coordinate of a product group on the image, the coordinate of the product group is a coordinate satisfying a condition within an area indicating the product group; recognize a position of the shelf label indicating information of a product from the image; determine that the shelf label is at a boundary between the first product and the second products, which are adjacent to each other; determine a first horizontal distance between a right end of the first product on a left side of the shelf label and a center of the shelf label, and a second horizontal distance between a left end of the second product on a right side of the shelf label and the center of the shelf label; detect an error in a positional relationship between the recognized first position of the first product and the recognized position of the shelf label based on the first horizontal distance and the second horizontal distance; and output the image on which a location including the first position of the first product related to the detected error is highlighted.
2. The information processing apparatus according to claim 1, wherein the processor is further configured to execute the instructions to: when the image is generated, or when the image is acquired, recognize the position of at least one product and the position of the shelf label.
3. The information processing apparatus according to claim 1, wherein the information processing apparatus is a portable terminal.
4. The information processing apparatus according to claim 3, wherein the processor is further configured to execute the instructions to: output the image on a display screen of the portable terminal.
5. The information processing apparatus according to claim 3, wherein the processor is further configured to execute the instructions to: generate the image by a capture unit of the portable terminal.
6. The information processing apparatus according to claim 1, wherein the processor is further configured to execute the instructions to: perform notification or displaying that there is an error, when there is the error in the relationship between the recognized position of the first product and the recognized position of the shelf label is detected.
7. The information processing apparatus according to claim 1, wherein the processor is further configured to execute the instructions to: perform notification or displaying that there is no error, when there is no error in the relationship between the recognized position of the first product and the recognized position of the shelf label is not detected.
8. A control method performed by a computer, the method comprising: recognizing a first position of a first product and a second position of a second product from an image on which products and a shelf label are imaged by obtaining a coordinate of a product group on the image, the coordinate of the product group is a coordinate satisfying a condition within an area indicating the product group;

recognizing a position of the shelf label indicating information of a product from the image; determining that the shelf label is at a boundary between the first product and the second products, which are adjacent to each other; determining a first horizontal distance between a right end of the first product on a left side of the shelf label and a center of the shelf label, and a second horizontal distance between a left end of the second product on a right side of the shelf label and the center of the shelf label; detecting an error in a positional relationship between the recognized first position of the first product and the recognized position of the shelf label based on the first horizontal distance and the second horizontal distance; and outputting the image on which a location including the first position of the first product related to the detected error is highlighted.

9. The control method according to claim 8, further comprising: by the computer, when the image is generated, or when the image is acquired, recognizing the position of at least one product and the position of the shelf label.

10. The control method according to claim 8, wherein the computer is a portable terminal.

11. The control method according to claim 10, further comprising: by the computer, outputting the image on a display screen of the portable terminal.

12. The control method according to claim 10, further comprising: by the computer, generating the image by a capture unit of the portable terminal.

13. The control method according to claim 8, further comprising: by the computer, performing notification or displaying that there is an error, when there is the error in the relationship between the recognized position of the first product and the recognized position of the shelf label is detected.

14. The control method according to claim 8, further comprising: by the computer, performing notification or displaying that there is no error, when there no error in the relationship between the recognized position of the product and the recognized position of the shelf label is not detected.

15. A non-transitory computer-readable storage medium storing a program causing a computer to: recognize a first position of a first product and a second position of a second product from an image on which products and a shelf label are imaged by obtaining a coordinate of a product group on the image, the coordinate of the product group is a coordinate satisfying a condition within an area indicating the product group; recognize a position of the shelf label indicating information of a product from the image; determine that the shelf label is at a boundary between the first product and the second products, which are adjacent to each other; determine a first horizontal distance between a right end of the first product on a left side of the shelf label and a center of the shelf label, and a second horizontal distance between a left end of the second product on a right side of the shelf label and the center of the shelf label; detect an error in a positional relationship between the recognized first position of the first product and the recognized position of the shelf label based on the first horizontal distance and the second horizontal distance; and output the image on which a location including the first position of the first product related to the detected error is highlighted.

16. The non-transitory computer-readable storage medium according to claim 15, wherein the program further causes the computer to: when the image is generated, or when the image is acquired, recognize the position of at least one product and the position of the shelf label.

17. The non-transitory computer-readable storage medium according to claim 15, wherein the computer is a portable terminal.

18. The non-transitory computer-readable storage medium according to claim 17, wherein the program further causes the computer to: output the image on a display screen of the portable terminal.

19. The non-transitory computer-readable storage medium according to claim 17, wherein the program further causes the computer to: generate the image by a capture unit of the portable terminal.

20. The non-transitory computer-readable storage medium according to claim 15, wherein the program further causes the computer to: perform notification or displaying that there is an error,

when there is the error in the relationship between the recognized position of the first product and the recognized position of the shelf label is detected.
